9. Two Stage Least Squares and Modern IV Estimators

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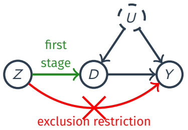
Outline

1. Two-Stage Least Squares
2. Modern IV via Machine Learning
3. Estimating IV: AJR Walkthrough + Modern IV
4. Notable IV Designs in Recent Empirical Work

Where are we? Where are we going?

- Last time:
  - **Instrumental variable** under noncompliance in randomized experiments.
    - The local ATE (or CACE): $\tau_{\text{LATE}} = \text{ITT}_{Y,\text{co}} = \frac{\text{ITT}_Y}{\text{ITT}_D}$
    - The Wald/IV estimator: $\widehat{\tau}_{\text{iv}} = \widehat{\text{ITT}}_Y / \widehat{\text{ITT}}_D$
    - Intent-to-treat analysis, compliance types, identification assumptions..



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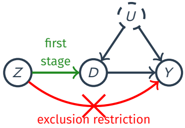

- Today:
  1. **Two-Stage Least Squares** (TSLS): estimands, weak IVs, multivalued D, general 2SLS.
  2. **Modern IV via Machine Learning**: PLIV (constant τ), DRIV (heterogeneous $\tau(\mathbf{X})$), DML.
  3. **Estimating IV in practice**: AJR walkthrough + modern IV in R.
  4. **Notable IV designs**: treatment assignments, regional characteristics, peers' environments, shift-share.

Source: *Chapter* 4 of Mostly Harmless Econometrics (Textbook 1) by J. Angrist & J. Pischke

1/ Two-Stage Least Squares

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- **Two-stage least squares** (TSLS) is the classical approach to IV which assumes two linear models:

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    - It's also exogenous for treatment uptake, so $\mathbb{E}[\eta_i | Z_i] = 0$.
- This implies the following CEF form for Y_i conditional on Z_i:

$$\mathbb{E}[Y_i | Z_i] = \alpha + \tau \mathbb{E}[D_i | Z_i] = \alpha + \tau \cdot (\gamma Z_i)$$

    - α is reused loosely (technically $\alpha + \tau\delta$ here); not the parameter of our interest.

TSLS Estimands

- Under the model, we have the following CEF: $\mathbb{E}[Y_i|Z_i] = \alpha + \tau \cdot (\gamma Z_i)$
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- TSLS estimator:
  - Estimate $\widehat{\gamma}$ from regression of treatment D_i on instrument Z_i
  - Estimate $\widehat{\tau}_{2SLS}$ as the slope of a regression of Y_i on $\widehat{\gamma} Z_i$.
    - $\widehat{D}_i = \widehat{\delta} + \widehat{\gamma} Z_i$; the Z-driven $\widehat{\gamma} Z_i$ is exogenous, uncorrelated with ε_i (constant $\widehat{\delta}$ doesn't affect the slope).
  - Under this model, $\widehat{\tau}_{2SLS} \xrightarrow{p} \tau$ (but don't use SEs from second stage; see MHE section 4.6.1. *2SLS Mistakes*)

Binary Treatment and Instrument

- Under binary treatment/instrument, TSLS estimand is the LATE:

$$\tau = \frac{\text{cov}(Y_i, Z_i)}{\text{cov}(D_i, Z_i)} = \frac{\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0]}{\mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0]} = \frac{\text{ITT}_Y}{\text{ITT}_D} = \tau_{\text{LATE}}$$

  - $\text{cov}(Y, Z)/\text{cov}(D, Z) = \beta_Y/\beta_D$ ($\mathbb{V}(Z)$ cancels, regardless of Z). For binary Z, OLS slope $\beta_\cdot = \mathbb{E}[\cdot|Z{=}1] - \mathbb{E}[\cdot|Z{=}0]$ (W4), giving $\text{ITT}_Y/\text{ITT}_D$.
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- And that the TSLS estimator is the Wald estimator:

$$\widehat{\tau}_{\text{2SLS}} = \frac{\widehat{\text{cov}}(Y_i, Z_i)}{\widehat{\text{cov}}(D_i, Z_i)} = \frac{\overline{Y}_1 - \overline{Y}_0}{\overline{D}_1 - \overline{D}_0} = \frac{\widehat{\text{ITT}}_Y}{\widehat{\text{ITT}}_D} = \widehat{\tau}_{\text{iv}}$$

$\rightsquigarrow$ Constant effects model is not required for TSLS in this setting.

TSLS with Covariates

- Adding covariates linearly:

$$Y_i = \alpha + \tau D_i + \mathbf{X}_i' \beta_y + \varepsilon_i$$
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- Why not Lin (W4) fully-interacted models ($\tau + \gamma'\widetilde{\mathbf{X}}_i$ varies) in IV?
  1. Each $D_i\widetilde{X}_{ij}$ is endogenous and requires its own instrument, e.g., $Z_i\widetilde{X}_{ij}$.
  2. Strong prediction of D by Z does not imply strong prediction of $D_i\widetilde{X}_{ij}$ by $Z_i\widetilde{X}_{ij}$.
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- $\rightsquigarrow$ **PLIV** (§2) inherits constant τ (ML-flexible $g(\mathbf{X})$, $m(\mathbf{X})$ only). **DRIV** (§2) relaxes to $\tau(\mathbf{X})$.

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- The bias of the Wald estimator:

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- **Relevant IV**: noise divided by signal $\longrightarrow 0$; **Irrelevant IV**: noise divided by noise does not.

What to Do About Weak Instruments?

- **Detect**: first-stage F-test on excluded instruments ($H_0 : \gamma = 0$).
  - $F \geq 10 \Rightarrow$ bias is small (Stock & Yogo 2005); still the dominant convention in empirical IV.
  - $F \geq 104.7$ for correct CI coverage in the *worst-case* DGP (Lee et al. 2022).
    - Lee et al. show the standard 1.96 gives true 95% coverage *only* at $F \geq 104.7$; below that, actual coverage drops monotonically. At the conventional $F = 10$, worst-case actual coverage is around **85%**.

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- **Weak-IV robust inference**: Anderson-Rubin (1949) test (simplified setting, binary Z, D).
  - **Trick**: rewrite $H_0 : \tau = \tau_0$ as $H_0 : \text{ITT}_Y - \tau_0\, \text{ITT}_D = 0$: **linearises the null**, no weak-IV pathology.
  - Under the null, asymptotically:

$$g(\tau_0) = \widehat{\text{ITT}}_Y - \tau_0\, \widehat{\text{ITT}}_D \sim N(0, \Omega(\tau_0))$$
$$\Omega(\tau_0) = \mathbb{V}[\widehat{\text{ITT}}_Y] + \tau_0^2\, \mathbb{V}[\widehat{\text{ITT}}_D] - 2\tau_0 \operatorname{cov}(\widehat{\text{ITT}}_Y, \widehat{\text{ITT}}_D)$$

  - AR statistic: $g(\tau_0)^2/\Omega(\tau_0) \sim \chi_1^2$ regardless of first-stage strength; CI by analytical inversion.

```
> library(ivmodel)
> m <- ivmodel(Y = y, D = D, Z = Z)
> m$AR    # AR test (F-stat), p-value, df, and 95% CI by inversion
```

  - Increasingly reported alongside the F-stat; handles weak-IV uncertainty correctly when first stage is borderline.

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- Can't identify the proportion of all compliance types here.
- Example: $K = 3 \rightsquigarrow 9$ principal strata
  - Affected: $(D_i(0), D_i(1)) \in \{(0,1), (0,2), (1,2)\}$
  - Unaffected: $(D_i(0), D_i(1)) \in \{(0,0), (1,1), (2,2)\}$
  - Negatively affected: $(D_i(0), D_i(1)) \in \{(1,0), (2,0), (2,1)\}$
  - Last ruled out by monotonicity.

TSLS with Multivalued Treatments

- Let $C_i = jk$ be an indicator for compliance type $D_i(1) = j$ and $D_i(0) = k$.
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2/ Modern IV via Machine Learning

Cf. Chernozhukov et al. (2018), *Double/Debiased Machine Learning*, Econometrics Journal

From Classical TSLS-with-X to Modern IV

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- Classical TSLS-with-$\mathbf{X}$ enters $\mathbf{X}$ linearly with a constant τ. Two limits given the above:
  - Linear/parametric $\mathbf{X}$ leaves nuisance specification fragile.
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- Both run on the **Double/Debiased ML** engine: cross-fitting + Neyman-orthogonal moment + ML for the three nuisance CEFs $\mathbb{E}[Y|\mathbf{X}], \mathbb{E}[D|\mathbf{X}], \mathbb{E}[Z|\mathbf{X}]$.

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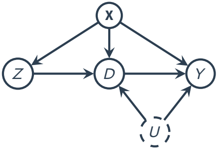

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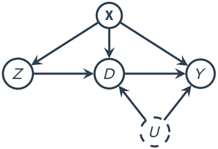

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  - When $\mathbf{X} \not\to Z$ (i.e., $Z \perp \mathbf{X}$), m_0 collapses to the constant $\mathbb{E}[Z]$ and ML has nothing to do; it becomes a real function of $\mathbf{X}$ only when $\mathbf{X}$ predicts Z.

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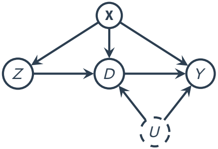

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- **Why naive plug-in fails**: ML regularization injects bias into $\widehat{\eta}$ that contaminates $\widehat{\tau}$. DML fixes this with two ideas:
  - **Neyman orthogonality**: moment is locally insensitive to nuisance errors at the truth.
  - **Cross-fitting**: fit $\widehat{\eta}$ on out-of-fold data, evaluate on held-out fold.

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  1. Partition into folds $\{I_k\}_{k=1}^K$. For each k, fit $\widehat{\ell}_{[k]}, \widehat{r}_{[k]}, \widehat{m}_{[k]}$ on data *outside* I_k.
  2. For $i \in I_k$: $\widetilde{Y}_i = Y_i - \widehat{\ell}_{[k]}(\mathbf{X}_i)$, $\widetilde{D}_i = D_i - \widehat{r}_{[k]}(\mathbf{X}_i)$, $\widetilde{Z}_i = Z_i - \widehat{m}_{[k]}(\mathbf{X}_i)$.
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- Software: `DoubleML` (R). Applied PLIV demo on AJR in §3.

DRIV: Letting τ Depend on X

- **One-symbol difference from PLIV**:

$$\text{PLIV:} \quad Y = \theta \cdot D + g(\mathbf{X}) + \varepsilon \quad (\text{constant } \theta)$$
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library(reticulate); econml_dr <- import("econml.iv.dr"); sk <- import("sklearn.ensemble")
est <- econml_dr$DRIV(
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- **R cousin**: `grf::instrumental_forest` (Causal Forest IV) – different moment (forest-weighted local 2SLS), same target $\tau(\mathbf{X})$, same conclusions in practice.

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- **Practical**: report R^2 for each candidate, rely on CVC-selected or stacked estimate. `ddml` R package implements all three.

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- Active research direction: combining ML's prediction strength with causal-graph reasoning so that the analyst (or the algorithm) does not silently introduce collider bias when assembling **X**. *Work in progress.*

3/ Estimating IV: AJR Walkthrough + Modern IV

From Theory to Practice: Classical TSLS Setup

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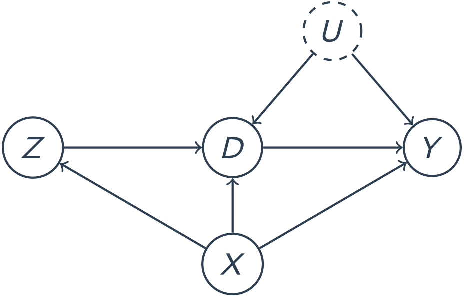

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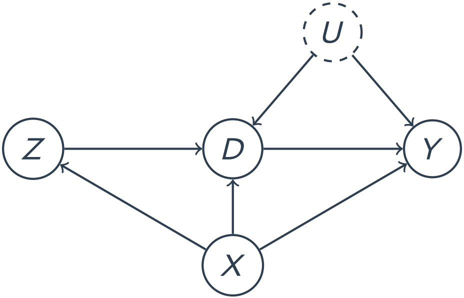

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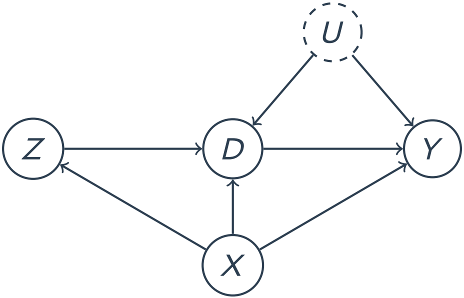

- $Z \to D$: first stage. $D \to Y$: causal effect of interest. U: unobserved confounder. X: observed covariates.
- Exclusion restriction: Z affects Y only through D (no direct $Z \to Y$ arrow).

Property Rights & Economic Development

Created using DALL·E 3 motivated by Ratna Sagar Shrestha

Recognize the person on the right?

- Q: Do property rights (i.e., institutions) promote economic development?
  - Famous paper on this: Acemoglu, Johnson, and Robinson (2001) AER
  - Relationship between strength of property rights in a country and GDP.

The AJR Data

Name	Description
shortnam	three-letter country code
africa	indicator for if the country is in Africa
asia	indicator for if country is in Asia
logem4	log mortality rates faced by European settlers (IV)
avexpr	strength of property rights (protection against expropriation)
logpgp95	log GDP per capita

```
> ajr <- read_csv("https://bit.ly/3RUJDWK"); ajr

# A tibble: 163 × 15
   shortnam africa lat_abst malfal94 avexpr logpgp95 logem4  asia yellow baseco leb95
   <chr>     <dbl>    <dbl>    <dbl>  <dbl>    <dbl>  <dbl> <dbl>  <dbl>  <dbl> <dbl>
 1 AFG           0   0.367   0.00372 NA       NA       4.54     1      0     NA  NA
 2 AGO           1   0.137   0.950     5.36     7.77   5.63     0      1      1  46.5
 3 ARE           0   0.267   0.0123    7.18     9.80  NA        1      0     NA  NA
 4 ARG           0   0.378   0         6.39     9.13   4.23     0      0      1  72.9
 5 ARM           0   0.444   0        NA        7.68  NA        1      0     NA  NA
 6 AUS           0   0.300   0         9.32     9.90   2.15     0      1      1  78.2
 7 AUT           0   0.524   0         9.73     9.97  NA        0      0     NA  NA
 8 AZE           0   0.448   0        NA        7.31  NA        1      0     NA  NA
 9 BDI           1   0.0367  0.950    NA        6.57   5.63     0      1     NA  NA
10 BEL           0   0.561   0         9.68     9.99  NA        0      0     NA  NA
# i 153 more rows
# i 4 more variables: imr95 <dbl>, meantemp <dbl>, lt100km <dbl>, latabs <dbl>
# i Use `print(n = ...)` to see more rows
```

In R: Example Code Using ajr Data

```
# Center (i.e., demeaning) the variables
> ajr <- ajr |>
    mutate(
      D_cnt = avexpr - mean(avexpr, na.rm = TRUE),
      Y_cnt = logpgp95 - mean(logpgp95, na.rm = TRUE),
      Z_cnt = logem4 - mean(logem4, na.rm = TRUE)
    ) |>
    na.omit()

# Compute the ITTs on D and on Y:
> ITT_D <- cov(ajr$Z_cnt, ajr$D_cnt)
> ITT_Y <- cov(ajr$Z_cnt, ajr$Y_cnt)
> wald_estimate <- ITT_Y / ITT_D; round(wald_estimate, digits = 4)
[1] 0.9242

# Same as reg Y~Z / reg D~Z:
> ITT_Y <- coef(lm(Y_cnt ~ Z_cnt, data = ajr))[[2]]
> ITT_D <- coef(lm(D_cnt ~ Z_cnt, data = ajr))[[2]]
> wald_estimate <- ITT_Y / ITT_D; round(wald_estimate, digits = 4)
[1] 0.9242
```

- The Wald estimate from manual calculation: 0.9242

In R: Example Code Using ajr Data

```
# Compare with ivreg
> ivreg_result <- AER::ivreg(Y_cnt ~ D_cnt | Z_cnt, data = ajr); summary(ivreg_result)

Call:
AER::ivreg(formula = Y_cnt ~ D_cnt | Z_cnt, data = ajr)

Residuals:
     Min       1Q   Median       3Q      Max
-2.40175 -0.54950  0.01792  0.68944  1.67361

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.031e-16  1.231e-01   0.000        1
D_cnt       9.242e-01  1.547e-01   5.974 1.59e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9458 on 57 degrees of freedom
Multiple R-Squared: 0.1107,        Adjusted R-squared: 0.09506
Wald test: 35.68 on 1 and 57 DF,  p-value: 1.589e-07

> round(coef(ivreg_result), digits = 4)
(Intercept)       D_cnt
     0.0000      0.9242
```

- The 2sls estimate from `AER::ivreg()`: 0.9242
  - Caveat: compute robust SE separately! (don't use SE from 2nd stage)

In R: Example Code Using ajr Data

```
# Estimate robust SE with estimatr::iv_robust()
> iv_rob <- estimatr::iv_robust(Y_cnt ~ D_cnt | Z_cnt, data = ajr, se_type = "HC2"); iv_rob
               Estimate Std. Error      t value     Pr(>|t|)   CI Lower CI Upper DF
(Intercept) 4.148820e-16  0.1234908 3.359619e-15 1.000000e+00 -0.2472860 0.247286 57
D_cnt       9.242173e-01  0.1791511 5.158871e+00 3.262823e-06  0.5654735 1.282961 57
```

- Robust SE with HC2 using `estimatr::iv_robust()`: 0.1791

```
> iv_rob_cov <- estimatr::iv_robust(Y_cnt ~ D_cnt + lat_abst + asia + africa |
                                     Z_cnt  + lat_abst + asia + africa, data = ajr,
                                    se_type = "HC2")
> var_labels <- c(
   "D_cnt" = "Avg. Expropriation Risk (D_cnt)", "lat_abst" = "Abs. Value of Latitude",
   "asia" = "Asian country", "africa" = "African country", "Z_cnt" = "log Mortality Rate (D_cnt)"
 )
```

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```

- When adjusting for covariates in TSLS, include them both in the 1st and 2nd stage.

IV Regression Table with modelsummary

```
> modelsummary::modelsummary(
  models = list(ivreg_result, iv_robust, iv_rob_cov),
  coef_map = var_labels, gof_map = c("nobs", "r.squared", "adj.r.squared"), stars = T,
  notes = "Note: See appendix for other model statistics.", output = "modelsummary_tab.tex")
```

	(1)	(2)	(3)
Avg. Expropriation Risk (D_cnt)	0.924***	0.924***	1.015**
	(0.155)	(0.179)	(0.351)
Abs. Value of Latitude			-1.596
			(1.529)
Asian country			-1.048*
			(0.425)
African country			-0.390
			(0.342)
Num.Obs.	59	59	59
R2	0.111	0.111	0.045
R2 Adj.	0.095	0.095	-0.026

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: See appendix for other model statistics.

Modern IV in R: DoubleML for PLIV

- Same partialling-out as Modern IV section, in ~10 R lines (cross-fitting built in):

```r
> library(DoubleML); library(mlr3); library(mlr3learners)
> # Build DoubleMLData (y = outcome, d = treatment, z = instrument, x = covariates)
> data <- DoubleMLData$new(df, y_col = "Y", d_cols = "D",
+                          z_cols = "Z", x_cols = X_names)
> # ML learners for the nuisance functions
> ml_l <- lrn("regr.ranger")   # E[Y | X]
> ml_m <- lrn("regr.ranger")   # E[Z | X]
> ml_r <- lrn("regr.ranger")   # E[D | X]
> # PLIV with 5-fold cross-fitting  (Neyman orthogonality + cross-fitting = DML)
> pliv <- DoubleMLPLIV$new(data, ml_l, ml_m, ml_r, n_folds = 5)
> pliv$fit(); pliv$summary()
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- **Bad controls**: DML does *not* pick confounders for you. ML selects by predictive power; colliders/mediators in $\mathbf{X} \rightsquigarrow$ biased $\widehat{\tau}$ (Hünermund, Louw & Caspi 2023, *JCI*). **DAG reasoning (ECI W6) still your job.**

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- Also: $\widehat{\tau}$ sensitive to learner / $\mathbf{X}$ choice (Ahrens et al. 2025 §5–6, by DML 2018 authors). Remedies proposed; revisit in **W15**.

What Does a BLI Ad Look Like?

- Blind Lead-In (BLI) ads are multi-stage interactive sliders. The product is revealed *only* on the last panel; users must advance through each panel themselves.

Panel 1	Panel 2	Panel 3	Reveal
Hook *narrative open* 100% of users	**Build** *narrative* drop-off begins	**Build** *complexity* ↑ cost rises	**Product + CTA** *shop now* ≈ 3.3% complete

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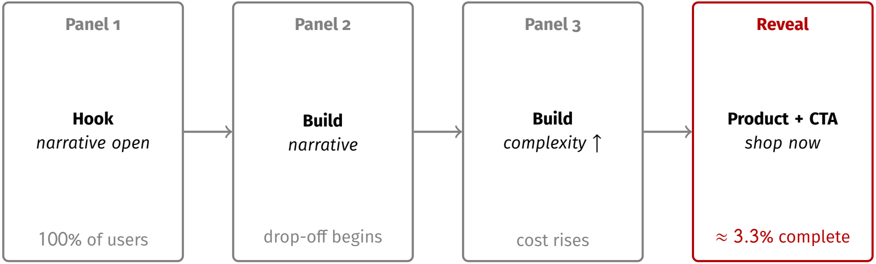

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- In passive media (TV, pre-roll), exposure $\Longrightarrow$ reveal automatically. In BLI, only $\approx$ 3.3% of exposed users reach the reveal.
- $\Longrightarrow$ The behavioral object of interest is D (completion), not Z (exposure). Standard ITT-on-Z averages over 96.7% never-takers.

Modern IV Demo: A Digital Advertising Experiment

- Online book retailer A/B-tests two ad formats ($n \approx 1.4$M user-impressions, 76 ad creatives).
  - Z: random assignment to interactive (Blind Lead-Ins, BLI) vs. static banner.
  - D: BLI *completion* (treated-side completion rate $\approx 3.32\%$; **one-sided noncompliance**: $D = 0$ for all $Z = 0$).
  - Y: 30-day purchase conversion (binary).

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- **Plan**: (i) classical pooled 2SLS LATE (`fixest`); (ii) DRIV via reticulate (`econml.LinearIntentToTreatDRIV`) for $\widehat{\theta}(\mathbf{X})$; (iii) Causal Forest IV (`grf::instrumental_forest`) for nonparametric $\widehat{\tau}(\mathbf{X})$. *Two distinct HTE-IV estimators on the same data; do they agree?*

Pooled 2SLS LATE: Classical Baseline

- 2SLS LATE on 30-day conversion (date FE, user covariates, cluster-robust SE at the user level):

```
> library(fixest)
> late <- feols(lngt_conv ~ ctrl_vars | date | tot ~ treatment,
+               data = d, cluster = ~ memno)
> summary(late)
TSLS estimation, Dep. Var.: lngt_conv
Observations: 1,408,121 / Fixed-effects: date (80)
Standard-errors: Clustered (memno)
        Estimate Std. Error t value Pr(>|t|)
fit_tot 0.002326   0.001210   1.922   0.0548 .
... (control coefficients omitted)
First-stage F (Kleibergen-Paap): ~1.5e4   # very strong instrument
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- **Weak-IV diagnostic**: first-stage $F = 10{,}007$ (Kleibergen-Paap, cluster-robust). Compliance is small (3.32%) but big-tech ad RCTs are typically large-N enough that the first stage is overwhelming; Stock-Yogo $F \geq 10$ **vastly** cleared.

DRIV via Reticulate (Syrgkanis et al. 2019)

- DRIV: heterogeneous LATE $\theta(X) = \alpha + \beta' X$ via cross-fitted nuisance + doubly-robust orthogonal score. Implementation: `econml.LinearIntentToTreatDRIV` (Python only). Call from R via `reticulate`:

```r
library(reticulate)
econml_iv <- import("econml.iv.dr"); sk <- import("sklearn.ensemble")
est <- econml_iv$LinearIntentToTreatDRIV(
  model_y_xw   = sk$RandomForestRegressor(n_estimators = 100L),
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  flexible_model_effect = "auto",  cv = 5L,  cov_clip = 0.005)
est$fit(Y = Y, T = D, Z = Z, X = X_paper, W = W_paper)
est$ate__inference()$summary_frame()
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- ATE across `cov_clip` sensitivity (key trimming knob for low-compliance cells):

`cov_clip`	ATE	s.e.	95% CI	Note
0.001	0.00197	0.00122	[−0.0004, +0.0044]	most permissive
0.005	0.00247	0.00107	[+0.0004, +0.0046]	**book default**
0.010	0.00209	0.00078	[+0.0006, +0.0036]	
0.050	0.00056	0.00019	[+0.0002, +0.0009]	over-trimmed

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- Linear projection $\beta'\mathbf{X}$ flags which features drive $\widehat{\theta}(\mathbf{X})$ heterogeneity. With face-stage decomposition (`first_has_face`, `middle_has_face`, `late_has_face`): `middle_has_face` is the only significant face term $\Rightarrow$ middle-panel face matters causally. Side-by-side with Causal Forest IV in the next slide.

From $\widehat{\tau}$ to $\widehat{\tau}(\mathbf{X})$: Causal Forest IV in R

- R-native HTE-IV (distinct from DRIV; §2): `grf::instrumental_forest` (Athey–Tibshirani–Wager 2019); **X** = 4 user + 4 creative features.

```
> library(grf)
> ivf <- instrumental_forest(
+   X = X_matrix, Y = Y, W = D, Z = Z,
+   num.trees = 4000, min.node.size = 100, honesty = TRUE)
> average_treatment_effect(ivf)               # forest-weighted ATE on BLI sample
> predict(ivf, X_new)$predictions             # tau-hat(X) at any covariate point
> best_linear_projection(ivf, A = X_matrix)   # which features drive HTE; valid SE
> test_calibration(ivf)                       # is HTE detectable above noise?
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- **Causal Forest ATE**: ≈ 0.17 **pp** on average (forest-weighted; conservative vs. pooled LATE because low-compliance cells are down-weighted).

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- Same target $\widehat{\tau}(\mathbf{X})$ as DRIV (§2), built from forest-weighted local 2SLS rather than the orthogonal-score moment.

What HTE-IV Reveals on the BLI Data

- **Pooled-LATE ladder converges**: 2SLS 0.00229, DML PLIV (Lasso) 0.00217, DML IIVM (RF) 0.00164 ($p = 0.049$), DRIV (econml, cov_clip=0.005) 0.00247. All in the [+322%, +484%] lift band $\Longrightarrow$ effect robust to ML, score, and functional-form choice.

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- **DRIV and Causal Forest IV both flag middle-panel face presence**:
  - `middle_has_face` (positive): Causal Forest BLP $\widehat{\beta} = +0.0144$ ($p = 0.008$); DRIV linear $\widehat{\beta} = +0.0118$ ($p = 0.014$). *Both significant.*
  - `late_has_face` (negative): Causal Forest $\widehat{\beta} = -0.0103$ ($p = 0.086$, marginal); DRIV $\widehat{\beta} = -0.005$ ($p = 0.33$, not significant). The forest picks up the late-stage penalty more sharply.
  - Engaged users (high recent visit-days) absorb ≈ 0.39 pp; disengaged ≈ 0.05 pp $\Longrightarrow$ near-8$\times$ heterogeneity gap pooled 2SLS hides.

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  - Engaged users (high recent visit-days) absorb ≈ 0.39 pp; disengaged ≈ 0.05 pp $\Rightarrow$ near-8$\times$ heterogeneity gap pooled 2SLS hides.
- **Mapping back to the paper**: H4 (face presence raises completion) confirmed at the *conversion-LATE* level. Stage asymmetry (middle dominant, late negative) is now *causally identified* on the downstream outcome, not just a descriptive completion-stage association.

Lift Translation: from $\widehat{\tau}$ to Industry Magnitude

- Gordon, Zettelmeyer, Bhargava & Chapsky 2019, *Marketing Science* eq. (8): lift = incremental conversion among treated, expressed as a percentage of the treated group's *counterfactual* (untreated) conversion.

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- BLI lift table (control conversion 0.051% as estimator of treated-counterfactual baseline):

Method	$\widehat{\theta}$	Compliers conv.	**Lift (%)**	NNE
2SLS pooled (paper)	0.00229	0.280%	+449%	13,200
DML PLIV (Lasso)	0.00217	0.268%	+426%	13,900
DML IIVM (RF)	0.00164	0.215%	+322%	18,400
DRIV (econml, cov_clip 0.005)	0.00247	0.298%	+484%	12,300

- Industry benchmark: typical online display-ad NNE is 50,000–200,000. BLI's 12,300–18,400 is order-of-magnitude better.

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- **Back-of-envelope**: 100M annual impressions $\times$ 3.32% completion $\times\,\widehat{\tau}_{\text{LATE}}$ (≈ 0.0023) $\approx$ 7,640 **extra purchases/year** causally attributable to BLI completion ($\approx$ 114M KRW at the retailer's average book price).

4/ Notable IV Designs in Recent Empirical Work

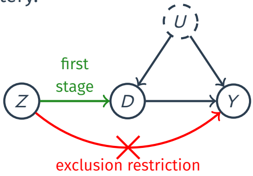
When you have a great idea for an IV but the first stage turns out to be not significant.

Source: Causal Inference for the Brave and True By Matheus Facure Alves

Treatment Assignments as IVs

Bloom et al. (*QJE* 2015)[1]

- Context: 9-month RCT in Ctrip's Shanghai call-centre; 249 volunteers entered the lottery.



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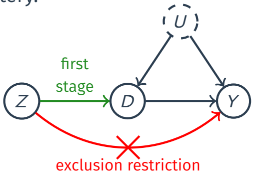

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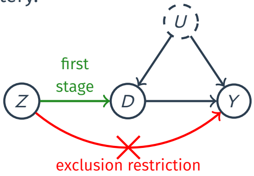

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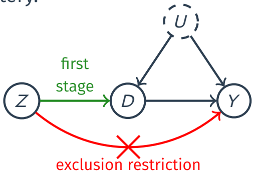

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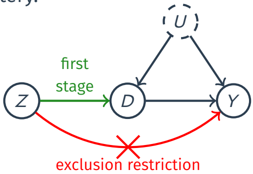

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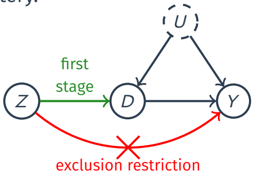

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- IV logic:
  1. Randomization: Balance tests on 18 baseline characteristics show no joint differences; p=0.466.
  2. Relevance: 80-90% of winners comply (1st-stage F ≈ 23)
  3. Exclusion: tasks, pay, IT identical – only location changes.
  4. Monotonicity: winners may or may not take WFH, losers cannot ⇝ $D_i(1) \geq D_i(0)$ trivially (one-sided non-compliance).

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Regional Characteristics as IVs

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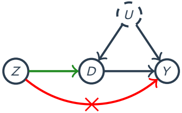

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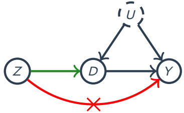

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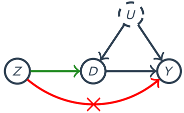

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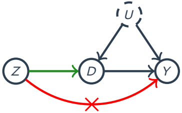

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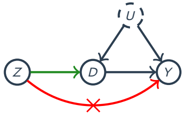

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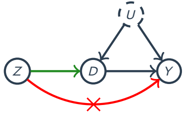

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- IV logic:
    1. Randomization: 90% of cell-towers built before 2010 and shows only weak correlations with 2010–15 population growth and store count, included in $\mathbf{X}_i$.
    2. Relevance: Each cell tower raises the log-odds of adoption by 0.0022 (z=3.7).
    3. Exclusion: controlled for ZIP income, store distance, and competitor presence to block direct demand channels.
    4. Monotonicity: Better signal only increases the prob. of downloading the app (no defiers).

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Peers' Environments as IVs

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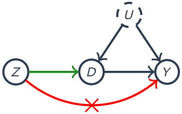

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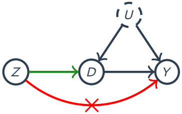

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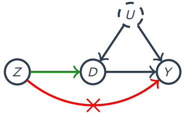

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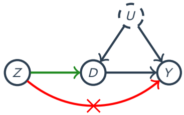

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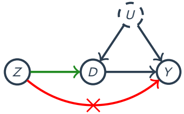

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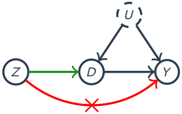

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- Identification issue: Homophily & shared shocks. Similar friends exercise together or face the same local weather.
- IV logic:
  1. Randomization: Use only friend pairs in different cities whose weather paths are uncorrelated; ego's own weather + date FE included.
  2. Relevance: first-stage F = 216–430 well above Stock-Yogo cutoff.
  3. Exclusion: Weather in friend's city is uncorrelated with ego's weather by construction.
  4. Monotonicity: bad weather never makes the friend run more.

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Onto the presentations & discussions!

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jaewon.yoo@iss.nthu.edu.tw
https://j1yoo.github.io/

Appendix

BLI Demo: Method-Comparison Table

Method	$\widehat{\theta}$	s.e.	95% CI	Notes
2SLS pooled (paper)	0.00229	0.00121	$[-0.00004, 0.00469]$	`fixest`, date FE, cluster=memno
DML PLIV (Lasso)	0.00217	0.00116	$[-0.00010, 0.00444]$	Book Ch 13.1 score, $n_{\text{rep}} = 5$
DML PLIV (RF, m=featureless)	0.00137	0.00190	$[-0.00235, 0.00509]$	Book Ch 13.1
DML IIVM (Lasso)	0.00217	0.00116	$[-0.00009, 0.00444]$	Book Ch 13.2
DML IIVM (RF)	0.00164	0.00083	$[+0.00001, 0.00328]$	**p = 0.049**
DRIV linear [cov_clip 0.001]	0.00197	0.00122	$[-0.00041, 0.00435]$	`econml.LinearIntentToTreatDRIV`
DRIV linear [cov_clip 0.005]	0.00247	0.00107	$[+0.00037, 0.00458]$	**book default**
DRIV linear [cov_clip 0.010]	0.00209	0.00078	$[+0.00057, 0.00362]$	

- All point estimates in $[0.00164, 0.00247]$, lift band $[+322\%, +484\%]$ over 0.051% control conversion.
- Working paper identification + inference still under revision; values shown are illustrative for the lecture demo.

BLI Demo: Subgroup CATE (engaged vs disengaged users)

Subgroup	n	CATE
Engaged (high recent visit-days)	686,903	0.00222
Disengaged (low recent visit-days)	721,218	0.00114
Age ≥ 30	932,236	0.00146
Age < 30	475,885	0.00207
`face_early` present	586,313	0.00235
`face_early` absent	821,808	0.00118

- Engaged $\times$ face-early users absorb *nearly all* the BLI completion lift; disengaged users with no early face: near-zero CATE.
- Source: `grf::instrumental_forest` subgroup CATE; trimmed cells with compliance $< 0.5\%$.

General 2SLS

- Linear model for each i:

$$Y_i = \mathbf{X}_i'\beta + \varepsilon_i$$

  - $\mathbf{X}_i$ is $k \times 1$ and now includes D_i and any pretreatment covariates.
  - Parts of $\mathbf{X}_i$ are endogenous so that $\mathbb{E}[\varepsilon_i \mid \mathbf{X}_i] \neq 0$
- Instruments $\mathbf{Z}_i$ that is $\ell \times 1$ vector such that $\mathbb{E}[\varepsilon_i \mid \mathbf{Z}_i] = 0$.
  - $\mathbf{Z}_i$ might include exogenous/pretreatment variables from $\mathbf{X}_i$ as well.
  - Rank condition: $\mathbb{E}[\mathbf{Z}_i\mathbf{Z}_i']$ and $\mathbb{E}[\mathbf{X}_i\mathbf{Z}_i']$ have full rank.
- Identification:
  - $k = \ell$: just-identified.
  - $k < \ell$: over-identified (can test the exclusion restriction, kinda)
  - $k > \ell$: unidentified (fails rank condition)

Nasty Matrix Algebra

- Projection matrix projects values of $\mathbf{X}_i$ onto $\mathbf{Z}_i$:

$$\mathbf{\Pi} = (\mathbb{E}[\mathbf{Z}_i\mathbf{Z}_i'])^{-1}\,\mathbb{E}[\mathbf{Z}_i\mathbf{X}_i'] \quad \text{(projection matrix, i.e., pi/}\mathbf{\Pi}\text{)}$$

$$\tilde{\mathbf{X}}_i = \mathbf{\Pi}'\mathbf{Z}_i \quad \text{(projected values)}$$

- To derive the 2SLS estimator, take the fitted values, $\mathbf{\Pi}'\mathbf{Z}_i$ and multiply both sides of the outcome equation by them:

$$\begin{aligned}
Y_i &= \mathbf{X}_i'\beta + \varepsilon_i \\
\mathbf{\Pi}'\mathbf{Z}_i Y_i &= \mathbf{\Pi}'\mathbf{Z}_i\mathbf{X}_i'\beta + \mathbf{\Pi}'\mathbf{Z}_i\varepsilon_i \\
\mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i Y_i] &= \mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i\mathbf{X}_i']\beta + \mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i\varepsilon_i] \\
\mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i Y_i] &= \mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i\mathbf{X}_i']\beta + \mathbf{\Pi}'\mathbb{E}[\mathbf{Z}_i\varepsilon_i] \\
\mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i Y_i] &= \mathbb{E}[\mathbf{\Pi}'\mathbf{Z}_i\mathbf{X}_i']\beta \\
\mathbb{E}[\tilde{\mathbf{X}}_i Y_i] &= \mathbb{E}[\tilde{\mathbf{X}}_i\mathbf{X}_i']\beta \\
\beta &= \left(\mathbb{E}[\tilde{\mathbf{X}}_i\mathbf{X}_i']\right)^{-1}\mathbb{E}[\tilde{\mathbf{X}}_i Y_i]
\end{aligned}$$

How to Estimate the Parameters

- Collect $\mathbf{X}_i$ into an $n \times k$ matrix $\mathbb{X} = (\mathbf{X}_1', \ldots, \mathbf{X}_n')$
- Collect $\mathbf{Z}_i$ into an $n \times \ell$ matrix $\mathbb{Z} = (\mathbf{Z}_1', \ldots, \mathbf{Z}_n')$

- In-sample projection matrix produces fitted values:

$$\widehat{\mathbb{X}} = \mathbb{Z}(\mathbb{Z}'\mathbb{Z})^{-1}\mathbb{Z}'\mathbb{X}$$

  - Fitted values of the regression of $\mathbb{X}$ on $\mathbb{Z}$.
  - Matrix party trick: $\mathbb{X}'\mathbb{Z}/n = (1/n)\sum_i^n \mathbf{X}_i\mathbf{Z}_i' \xrightarrow{p} \mathbb{E}[\mathbf{X}_i\mathbf{Z}_i']$.

- Take the population formula for the parameters:

$$\beta = \left(\mathbb{E}[\tilde{\mathbf{X}}_i\mathbf{X}_i']\right)^{-1}\mathbb{E}[\tilde{\mathbf{X}}_i Y_i]$$

- And plug in the sample values (the n cancels out):

$$\widehat{\beta}_{2SLS} = (\widehat{\mathbb{X}}'\mathbb{X})^{-1}\widehat{\mathbb{X}}'\mathbf{y} \xrightarrow{p} \beta$$

- This is how R/`Stata` estimate the 2SLS parameters.

Asymptotic Variance for 2SLS

- We can write the centered, normalized TSLS estimator as:

$$\sqrt{n}\left(\widehat{\beta}_{\text{2SLS}} - \beta\right) \;=\; \underbrace{\left(n^{-1}\sum_i \widehat{\mathbf{X}}_i\widehat{\mathbf{X}}_i'\right)^{-1}}_{\xrightarrow{p}\ (\mathbb{E}[\widehat{\mathbf{X}}_i\widehat{\mathbf{X}}_i'])^{-1}} \; \underbrace{\left(n^{-1/2}\sum_i \widehat{\mathbf{X}}_i\varepsilon_i\right)}_{\xrightarrow{d}\ N\left(0,\,\mathbb{E}[\widehat{\mathbf{X}}_i'\varepsilon_i'\varepsilon_i\widehat{\mathbf{X}}_i]\right)}$$

- Thus, $\sqrt{n}\left(\widehat{\beta}_{\text{2SLS}} - \beta\right)$ has asymptotic variance:

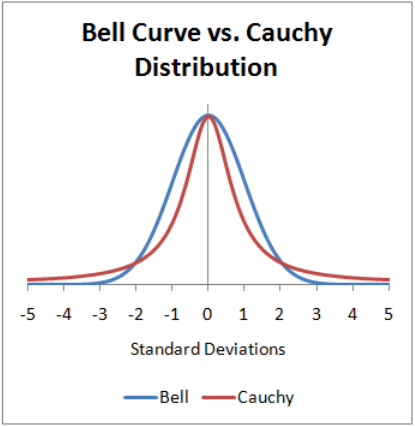
$$\left(\mathbb{E}[\widehat{\mathbf{X}}_i\widehat{\mathbf{X}}_i']\right)^{-1}\mathbb{E}[\widehat{\mathbf{X}}_i'\varepsilon_i'\varepsilon_i\widehat{\mathbf{X}}_i]\left(\mathbb{E}[\widehat{\mathbf{X}}_i\widehat{\mathbf{X}}_i']\right)^{-1}$$

- **Robust 2SLS variance estimator** with residuals $\widehat{u}_i = Y_i - \mathbf{X}_i'\widehat{\beta}$:

$$\widehat{\text{Var}}\left(\widehat{\beta}_{\text{2SLS}}\right) \;=\; (\widehat{\mathbb{X}}'\widehat{\mathbb{X}})^{-1}\left(\sum_i \widehat{u}_i^2\widehat{\mathbf{X}}_i\widehat{\mathbf{X}}_i'\right)(\widehat{\mathbb{X}}'\widehat{\mathbb{X}})^{-1}$$

- HC2, clustering, and autocorrelation versions exist.

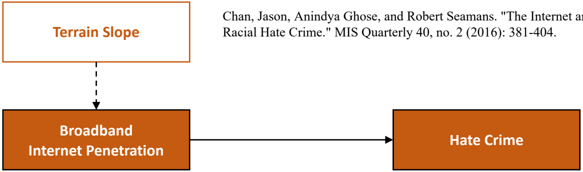
Cauchy vs. Normal Distribution



Source: https://stats.stackexchange.com/questions/36027/why-does-the-cauchy-distribution-have-no-mean

Regional Characteristics as IVs

- Chan, Ghose, Seamans, MISQ, 2016:

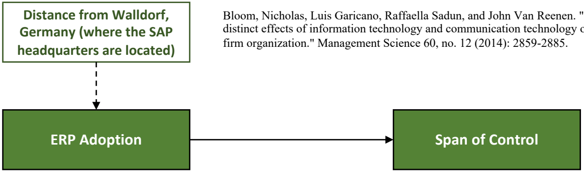


Chan, Jason, Anindya Ghose, and Robert Seamans. "The Internet and Racial Hate Crime." MIS Quarterly 40, no. 2 (2016): 381-404.

  - Observational study using *Hate Crime Statistics* from FBI.
    - RQ: Does the spread of Internet increase racial hate crime?
  - Issue? Confounding
  - IV? terrain slope/steepness (i.e., how many hills in a given region?)

Geographical Proximity-Based IVs

- RQ: Does the adoption of an ERP system impact plant manager autonomy (i.e., span of control)?
  - Span of control: the number of employees managed by supervisors or managers in an organization (high SoC = centralized).

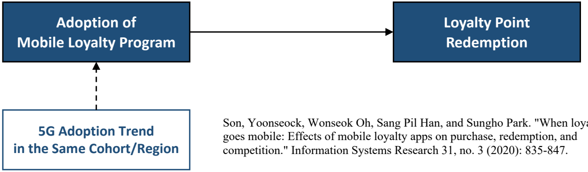


Bloom, Nicholas, Luis Garicano, Raffaella Sadun, and John Van Reenen. "The distinct effects of information technology and communication technology on firm organization." Management Science 60, no. 12 (2014): 2859-2885.

  - Observational study: the CEP management and organization survey and the Harte-Hanks ICT panel (Bloom, Garicano, Sadun, Van Reenen, MgmtSci, 2014).
  - IV: Distance from the ERP market leader (i.e., SAP) w/ 25%+ market share ⇝ likely more established connections with SAP (German firms vs. French, England firms).

Macro/Cohort Trends as IVs

- RQ: How does loyalty app adoption affect customer point redemption behavior?

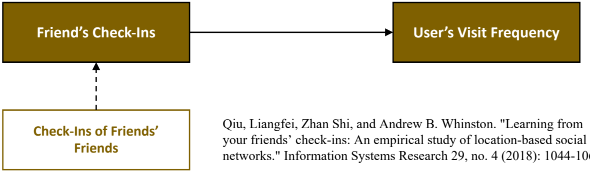


Son, Yoonseock, Wonseok Oh, Sang Pil Han, and Sungho Park. "When loyalty goes mobile: Effects of mobile loyalty apps on purchase, redemption, and competition." Information Systems Research 31, no. 3 (2020): 835-847.

  - Observational study: data on loyalty app adoption status, loyalty point redemption patterns, and purchase behaviors in multivendor loyalty program (MVLP) context (Son, Oh, Han, Park, ISR, 2020).
  - IV: 5G adoption rate in the same cohort (e.g., age group).

Peers' Environments as IVs

- Qiu, Shi, Whinston, ISR, 2018:



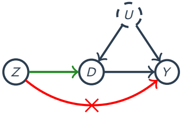
Qiu, Liangfei, Zhan Shi, and Andrew B. Whinston. "Learning from your friends' check-ins: An empirical study of location-based social networks." Information Systems Research 29, no. 4 (2018): 1044-1061.

  - Observational study using data on restaurant check-in information and the users' social network ties from a major SNS in China.
    - RQ: Is there observational learning/herding effect for restaurant discovery?
  - Issue? Homophily.
  - IV: Check-in activities of friend's friends.

Bartik (Shift-Share) IVs

Barron et al. (*Marketing Sci.* 2020)[4]

- Context: Panel of 43,000 ZIP codes (100 largest U.S. metro areas), monthly 2011-2016. Airbnb listings scraped; Zillow rent & price indices matched at ZIP-month.



Instrument Z: Google-Trends "Airbnb" (global shock) × 2010 ZIP "touristiness" (# food/lodging firms)

Treatment D: $\ln(1 + \text{Airbnb listings})$ in ZIP-month.

Outcome Y: Zillow rent index, house price index , price-to-rent.

- Identification issue: Hot ZIPs attract both Airbnb supply and rising housing costs $\rightsquigarrow$ upward bias.
- IV logic:
  1. Conditional exogeneity: 2010 touristiness fixed before Airbnb's national growth; Google-Trends shock is global, not driven by ZIP-specific factors. **(Hardest assumption for Bartik IVs; defended in detail next slide.)**
  2. Relevance: first-stage F = 650–820 well above Stock-Yogo cutoff.
  3. Exclusion: global search shocks unrelated to local housing; touristiness fixed in 2010. City×month FE + placebo ZIPs (no listings) show no direct price effect.
  4. Monotonicity: more global Airbnb awareness cannot lower listings in touristy ZIPs.

[4] Barron, K., Kung, E., & Proserpio, D. "The Effect of Home-Sharing on House Prices and Rents" Marketing Sci. 2020. 40(2): 283–304.

Bartik (Shift–Share) IVs

Barron et al. (*Marketing Sci.* 2020)

- IV logic 1 (continued): defending **conditional exogeneity**, the hardest assumption for Bartik IVs:
    - Parallel pre-trends: prior to 2012, rents and prices evolve identically across touristiness quartiles (Fig. 5) $\rightsquigarrow$ no pre-existing divergence.
    - Touristiness $\times$ time test: adding a ZIP-specific trend term $(h_{i,2010} \times t)$ yields an insignificant coefficient; IV effect unchanged.
    - Rich time-varying controls: results robust after adding ZIP income, population, hotel rooms, TripAdvisor reviews, and airport arrivals $\rightsquigarrow$ IV not picking up gentrification or tourism shocks.